NukMiris v0.8.1 — Independent Greenfield Architecture Challenge

Deliverable 1 of 5 — Independent Requirements Assessment

Independent architect submission · 26 October 2026 · Design only: no code, schema, configuration or provider asset was touched.

1. Basis, scope and independence

- **Inputs read:** the assignment brief and `01-Requirements-Baseline.md` (accepted v0.8.1, tag `95decc8…`). I did not read documents 02–05, the application source or the tests. Where the baseline is silent or tense, I list the gap below instead of resolving it from code.
- **IDs** (S01, W01–W09, C01–C09, R01–R12, I01–I13, O01–O04, Q01, U-*) are the baseline's.
- **No provider was contacted.** Claims about Google or Meta behaviour that I cannot verify from the baseline are marked **[verify S-n]** and collected in §7 as spikes that must pass before the design is committed.
- **Numbers I propose** (§6) are assumptions for owner approval. The baseline supplies no commercial targets and I do not treat mine as requirements.

2. Summary of the assessment

The baseline is unusually good input. The A–E classification separates product behaviour from mechanism, and the B-class invariants (I01–I13) are what actually decide the architecture. Four properties drive almost everything:

1. **Google Calendar is the truth for appointments, but no transaction spans Google and the platform.** Every booking, move, cancellation and attendance write is a two-system update with an unavoidable uncertainty window.
2. **WhatsApp sends are not idempotent** (no provider-side deduplication is described), so an unknown send outcome can never be resolved by retrying.
3. **Restart and multi-instance correctness are mandatory** (I04, I05, I06, I12), so in-process locks and in-memory queues are not acceptable coordination mechanisms.
4. **Backups cannot undo external effects** (O03). Recovery therefore needs inputs from outside the database.

I found 17 points where the baseline is silent, ambiguous or internally tense in ways that change design. Eight are high impact and need an owner decision before build: **F-01, F-02, F-03, F-04, F-05, F-06, F-12, F-13**. None blocks the architecture, but each fixes an assumption the data model or recovery design leans on.

3. Load-bearing requirements and what they force

Requirement	What it forces in any compliant design
I04, I05, I12 (accept before acknowledging; ordered; survive crashes)	Accepted input, state changes and queued replies live in one transactional store so that "accepted" and "processed" cannot diverge.
Section 2, C03, C06 (Google is truth; failure is not "empty")	An appointment projection is only a cache. Every adapter needs an explicit *unavailable* result distinct from *empty*.
I07, I09, Section 4 (definite vs uncertain outcomes)	A tri-state outcome type (applied, definitely rejected, uncertain) on every external call, plus an operation-specific read-only reconciliation procedure.
I06 (cooperating operations cannot finalize conflicting intervals; stale workers cannot release successors)	Database-level serialization with fencing. Process-local locks fail the moment two instances run.
I01, W02 (tenant authority from the receiving channel)	Tenant is derived once, from authenticated provider metadata, then enforced structurally at the data layer, not by repeated application checks.
R02, R03, I10 (instants for offsets; 15-minute late allowance; no replay after downtime)	A durable schedule of occurrences with absolute due instants and an expiry. A timer or queue delay alone is insufficient.
R09–R11 (permission and template readiness gate live sends)	Permission evidence as append-only data and template readiness as data, evaluated fail-closed at the moment of each send.
Section 3 (15-minute transient views; stale and guessed actions refused)	Offers are server-side records with opaque identifiers. Encoding state in button payloads would not meet I03.

Requirement	What it forces in any compliant design
O03, I12 and the restore scenario	Disaster recovery must treat provider state (Calendar events, delivered messages) as out-of-band facts that reconciliation reads back.

4. Findings: ambiguities, gaps and tensions

Each finding gives what is unclear, why it matters, the assumption I design to, and whether the owner must decide.

F-01 · Calendar capacity is one appointment at a time (High). The baseline specifies one Calendar, business-wide duration and no staff or resource concept, but never states capacity. *Why it matters:* capacity 1 allows a database exclusion constraint to serialize bookings. Capacity N or per-staff resources need a different concurrency model. *Assumption:* exactly one concurrent appointment per Calendar; multi-staff is E-class and would be modelled later as extra Calendars or resources. *Owner:* confirm.

F-02 · WhatsApp account and channel ownership (High). Tenant authority is "the authenticated receiving business channel", and each business uses its own approved templates and "authorized WABA/sending number" (R10). The baseline does not say who owns the WABA (business or platform acting as a technology provider), how numbers are onboarded, how credentials are issued, or who bears messaging cost. *Why it matters:* it fixes the credential shape (per-business token vs partner token), the blast radius of a stolen credential and the template-provisioning workload (several templates per business). *Assumption:* one phone number per business, mapped to exactly one active business by provider phone-number identifier; credentials held per business; onboarding stays an operator process. *Owner:* decide ownership and billing before the first paying tenant.

F-03 · "Do not repeat blindly" versus idempotent Calendar inserts (High). Section 4 says provider "not found" cannot prove a late insert never occurred, so the write must not be repeated blindly. As far as I know, Google Calendar accepts a client-chosen event identifier and rejects duplicates **[verify S-1]**. A repeat with the *same* identifier, followed by a read, is idempotent and therefore not blind. *Why it matters:* it turns uncertain bookings from "operator reconciliation" into automatic safe resolution. *Assumption:* the

prohibition targets non-idempotent repeats; a same-identifier retry with read verification is permitted. If S-1 fails, uncertain inserts stay uncertain until read-only proof or an operator decision. *Owner:* confirm the interpretation.

F-04 · What "Calendar ownership metadata" means; one Calendar, one business (High). The baseline does not define what marks an event as NukMiris-managed or whether two businesses may map the same Google Calendar. *Why it matters:* tenant isolation, manual-change detection (R08) and post-restore reconciliation all depend on it. *Assumption:* managed events carry private extended properties (business, customer, operation and schedule identity). Managed means those properties are present *and* match the acting business. A Calendar may map to at most one active business, enforced by a uniqueness constraint. Any mismatch fails safe. *Owner:* confirm that shared Calendars are out of scope.

F-05 · How long an uncertain outgoing message holds later replies (High). I05, I09 and I13 say unknown sends are never blindly resent and expired output must not block fresh work indefinitely. The baseline does not say how long an *unexpired* uncertain reply holds later replies in the same conversation, nor how long to wait for receipt evidence. *Assumption:* strict in-order delivery with a bounded receipt-wait (default 5 minutes, configurable). After that the conversation proceeds and the attempt stays `uncertain` in the review queue. *Owner:* approve the default.

F-06 · Customer phone identity (High). The baseline forbids asking for a phone already supplied and shows an "international" number in Calendar text, but does not define the canonical key. WhatsApp identifiers and dialled formats differ in some Latin American countries (Mexico and Argentina are the usual examples) **[verify S-9]**. *Why it matters:* it decides scoped-identity uniqueness, send-to formatting and Calendar text. *Assumption:* the identity key is the provider-supplied `wa_id` exactly as received, per business. A display form derived by a phone library is never used as a key. *Owner:* no decision needed, but the S-9 result may change formatting rules.

F-07 · Ordering across independent streams (Medium). Ordering is required per conversation for inbound processing and reply groups. The baseline does not relate a conversation reply, a late operation result and a reminder to the same customer, nor say whether a reminder waits while the customer is mid-booking. *Assumption:* operation results join the conversation's ordered outbox. Reminders are an independent stream that never waits on browsing state. Interleaving a reminder with a booking dialogue is accepted.

F-08 · Results that resolve after the service window (Medium). An uncertain operation may resolve hours later. Ordinary replies are limited to the 24-hour window (I13) and the only templates are reminders. The baseline does not say what the customer is told if the window has closed. *Assumption:* no proactive message; the recorded result is returned the next time the customer writes, and the operator can contact the business out of band. *Owner:* accept.

F-09 · No hold during name entry (Medium). Browsing never reserves and execution rechecks after name entry (Section 3), so a new customer can finish name entry and then lose the slot. *Assumption:* parity with the baseline: reserve only at submission; on a failed recheck, return fresh nearby options. A short TTL hold would be an additive change if the product wants it.

F-10 · Configuration changes while work is outstanding (Medium). Hours, duration, buffer and capability flags can change while offers, operations and reminders exist. The baseline does not say which version governs a pending operation. *Assumption:* configuration is versioned; each operation records the version it used; existing appointments keep the interval recorded in Calendar; an offer is invalidated if an availability-relevant field changed.

F-11 · Human handoff has no human (Medium, product and operations). Handoff records are stored but nobody is notified (Section 2). A customer in human mode gets no automation until they send Menu. This is a perceived-reliability risk, not a correctness defect. *Assumption:* the design stores handoffs and exposes a read-only operator queue with an oldest-unattended metric. It adds no customer-facing behaviour. *Owner:* be aware; I recommend at least an operator alert before commercial launch.

F-12 · Jurisdiction, privacy and retention (High for commercial use). U-OPS leaves retention, residency and deletion open. The target market spans countries with distinct data-protection regimes (for example Mexico, Brazil, Argentina, Colombia and Chile). Customer names and phones also live in the business's Google Calendar, which NukMiris does not own. *Why it matters:* it decides region, retention jobs, erasure mechanics, how consent evidence is stored and how diagnostics are redacted. *Assumption:* minimal PII, per-data-class retention that is configurable, per-business export and deletion capability, and protected credentials. Region is fixed after the owner decides. *Owner:* required before a pilot with real customers.

F-13 · Provider standing and quotas (High). The baseline omits launch prerequisites that can fail independently of code: Google OAuth consent-screen publishing and verification (refresh tokens for apps left in testing may expire quickly **[verify S-5]**), Meta business

verification, display-name approval, per-WABA template approval, and messaging limits and quality ratings. *Assumption:* these are tracked as launch prerequisites with named owners, and the design exposes credential, channel and template health per business.

F-14 · Template lifecycle during live operation (Medium). R10 blocks sending when a template is paused, disabled or rejected, but not how NukMiris learns of a change between checks. *Assumption:* subscribe to template status and quality notifications **[verify S-7]**, store them as data, re-verify at send time (Meta remains final authority), alert the operator, and block reminders fail-closed.

F-15 · Time-zone rule changes and DST policy (Medium). U-DST leaves ambiguous (repeated) wall-clock times open, and the baseline says nothing about governments changing time-zone rules after an appointment is booked. *Assumption:* store the instant plus the business zone; reject nonexistent times; exclude repeated wall-clock starts from offers (conservative); keep time-zone data updatable without a code release.

F-16 · Operator model (Medium). "Authorized platform operator" has no authentication, authorization, headcount, offboarding or separation-of-duties specification, yet operators can retry uncertain messages, record consent evidence and restore backups. *Assumption:* one to three operators; single sign-on with MFA; read-only, support and admin roles; every mutating action requires a reason and is audited. Consent evidence recorded by an operator is attributable, but its truth is an off-platform business process.

F-17 · Customer-visible copy as a contract (Low). Some strings are exact (the no-appointments message, reminder bodies, label format) and others are only described. *Assumption:* all customer copy lives in a versioned catalog. The exactly-specified strings are snapshot-tested.

5. Disposition of the baseline's own open items

ID	My proposed resolution	Needs approval
U-NAME / U-COMMAND	Keep the stated validation as a pure, replaceable function. During name entry only the exact global `menu` command overrides; other navigation words are treated as names if they validate.	Yes

ID	My proposed resolution	Needs approval
U-DST	Reject nonexistent local times. Exclude repeated local times from offers. Preserve instants everywhere.	Yes
U-EXTERNAL	App-level concurrency safety via database constraint plus a fresh Calendar read before write. After each write, an overlap audit *flags* external overlaps to a review queue and never auto-cancels.	Yes (commercial policy)
U-REPLAY	Separate the *recorded outcome* of an operation from *current appointment state*. A repeat returns the outcome labelled as historical and adds current state when Google is readable.	Yes (wording)
U-ACTIVATION	Conservative parity: compare intent-creation time with activation. Store both intent and commit timestamps so either policy applies without migration.	Yes
U-GENERIC	Target a generic "Sí" only when exactly one dispatched, unresolved request exists across the customer's current upcoming schedules in that business. Otherwise ask for the specific button. Requests on cancelled, moved or past schedules cannot be targeted, so no arbitrary age cutoff is needed.	Yes
U-RETRY	Operator-only, scoped, audited retry with a reason and a duplication warning, recorded as a new linked attempt. Never automatic.	No (parity)
U-OPS	Proposed targets in §6; hosting recommendation in Deliverable 2.	Yes

6. Proposed targets (assumptions exposed for owner approval)

The baseline asks that proposed targets be exposed as assumptions. These are *my* proposals, chosen to size the design, not requirements.

ID	Proposed target	Used to decide
T-1	Design envelope of up to 500 businesses, roughly 5,000 appointments per day platform-wide and about 100,000 inbound provider events per day, with 10× headroom before any architectural change.	Single Postgres primary; no broker; no sharding.
T-2	Webhook acknowledgement p99 under 1 second. Acknowledgement only after durable commit. Monthly ingress availability 99.5%, with provider retries covering short outages **[verify S-7]**.	Process topology; HA database.
T-3	Customer-visible latency: browse step p95 under 4 seconds; booking result p95 under 8 seconds (Calendar-bound).	Availability-search budget and caching.
T-4	RPO at most 5 minutes; RTO at most 2 hours for full restore and 15 minutes for automatic failover.	Backup and DR design.
T-5	At least 99% of eligible reminder occurrences submitted inside the [due, due+15 min) window; zero duplicate reminder submissions by construction.	Scheduler polling cadence and alerting.
T-6	Uncertain operations resolved or escalated to review within 15 minutes (settle window 10 minutes).	Reconciler cadence.
T-7	Retention (provisional): raw provider payload archive 30 days; all other classes pending legal input.	Archive and purge jobs.

7. External-behaviour assumptions to verify before committing

The design leans on provider behaviours I could not confirm from here. Each is a spike with an explicit fallback.

ID	What must be true	If false
S-1	Google Calendar `events.insert` accepts a client-supplied event ID; a duplicate returns a conflict; a deleted event's ID stays unusable; a read right after a successful insert is consistent.	Fall back to read-only proof plus operator resolution (see F-03).
S-2	`If-Match` / ETag preconditions on patch and delete return 412 on mismatch.	Replace with read-compare-write inside the operation lease and widen the post-write audit.
S-3	`events.list` supports private-extended-property filtering together with `updatedMin`, `showDeleted` and `fields`.	Fetch broader windows and filter locally.
S-4	`freebusy.query` handles all-day, tentative and declined events as the baseline requires; per-calendar errors are reported distinctly; range and quota limits are known.	Use `events.list` and compute busy time locally.
S-5	Least-privilege OAuth scope that permits create, patch, delete and free/busy; verification requirements; refresh-token lifetime in production; `invalid_grant` taxonomy; PKCE support.	Accept a broader scope with documented justification.
S-6	An opaque caller-supplied value on outgoing messages is echoed in status notifications; no send-side idempotency key exists; error codes can be split into definite and uncertain classes.	Correlate by recipient, time window and provider message ID only; treat more outcomes as uncertain.

ID	What must be true	If false
S-7	Webhook retry schedule and duration; status identifiers; quoted-message context ID; template-status and quality notifications; signature over raw bytes.	Poll template status; tighten durable-ingress alerting.
S-8	Utility template TTL ranges and the practical rules of the 24-hour window for interactive messages.	Adjust reminder timing and reply expiry.
S-9	Phone identifier formats in target countries and the correct send-to format.	Per-country normalization table.
S-10	Postgres: exclusion constraint combining range overlap with a *not-equal* key (via `btree_gist`); row-level security with pooled connections; claim-query performance on the chosen managed service.	Use advisory locks plus a unique index on discretized slots.
S-11	Time-zone library behaviour for nonexistent and repeated local times; tzdata update path.	Pin and test a different library.
S-12	Managed-Postgres point-in-time restore granularity and restore duration meet T-4.	Revise T-4 or change provider.

8. Scope decisions I am making explicitly

- **One durable mode.** The baseline calls non-durable mode non-commercial and weaker. The greenfield system has no such mode, so there is no downgrade path to guard (O04).
- **Tenant resolution only from the receiving channel.** No test-only sender-to-business routing in the product code (D-class).
- **No LLM in the control path.** Date and intent recognition remain bounded, deterministic grammar, as specified.
- **No dashboard, billing or self-service onboarding** (E-class). Operator tooling is a CLI and private admin API.
- **No migration** of POC data or pre-v0.8.1 states.

- **"No extra confirmation" is respected.** An acknowledgement that a request is being processed is not a confirmation question.

9. Where this assessment may be wrong

- **I have not seen the source.** The baseline warns that some behaviours are source-only quirks. I may have missed a subtle decision encoded in tests.
- **F-03 rests on one provider feature.** If client-assigned event IDs do not behave as I expect, recovery gets more manual and the operator workload rises.
- **F-04 and F-01 may be too restrictive.** A pilot customer with two practitioners on one Calendar would break the capacity-1 assumption.
- **My targets are guesses.** T-1 through T-7 should be replaced by owner numbers when they exist.
- **Some findings are risk disclosures, not defects.** F-11 and F-13 describe commercial exposure the architecture can only partly mitigate.